TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 14

5-MINUTE REVISION

Large Sample Tests

The Central Limit Theorem and Z tests for proportions, means and standard deviations.

Textbook authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · Chapter revision

KAU Me-LON · www.agrinoteskau.com · Dept. of Agricultural Statistics, College of Agriculture, Vellayani

THE IDEA TO KEEP

Standardise the estimate using its standard error.

REMEMBER

Check independence and the relevant large-sample approximation before applying a z test.

APPLY IT

For a population proportion test, check expected success and failure counts using the null proportion.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 Central Limit Theorem

As n increases

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

- Large sample: generally $n \geq 30$.
- Needs **independent** observations and **finite variance**.
- **Standard error** = SD of a sampling distribution; for $\bar{x}$ it is $\sigma/\sqrt{n}$, which shrinks as n grows.

2 Z test decision rules

Test	Reject H_0 if	5%	1%
Two-tailed	$\lvert Z\rvert > Z_{\alpha/2}$	1.960	2.576
Right-tailed	$Z > Z_{\alpha}$	1.645	2.326
Left-tailed	$Z < -Z_{\alpha}$	−1.645	−2.326

State H_0 about the **population parameter**; compute Z from **sample statistics**.

3 Proportions

Single proportion, $H_0 : P = P_0$

$$Z = \frac{p - P_0}{\sqrt{P_0 Q_0/n}}, \quad Q_0 = 1 - P_0$$

Two proportions, $H_0 : P_1 = P_2$

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

Pooled $\hat{P} = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$, $\hat{Q} = 1 - \hat{P}$. Needs enough expected successes **and failures**.

4 Single mean, $H_0 : \mu = \mu_0$

σ known

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

σ unknown

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

With a large sample, s is a good substitute for σ.

5 Two means, $H_0 : \mu_1 = \mu_2$

Equal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

Pooled σ if unknown

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}}$$

Unequal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

For independent samples, the variance of a difference is the **sum** of the variances.

6 Two SDs

Approximate test

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1}+\frac{s_2^2}{2n_2}}}$$

Under normal-population assumptions.

7 Choosing the Z test

Question	H_0	Standard error uses
Is the germination rate 80%?	$P = P_0$	P_0 (hypothesised)
Do two districts have the same adoption rate?	$P_1 = P_2$	Pooled $\hat{P}$
Is mean yield 40 q/ha?	$\mu = \mu_0$	σ or s
Do two varieties yield the same?	$\mu_1 = \mu_2$	Common or separate variances

! Common mistakes

- ✗ Using p instead of P_0 in the single-proportion SE.
- ✗ Ignoring the sign of Z in one-tailed tests.
- ✗ Using 1.96 for a one-tailed 5% test (it's 1.645).

? Fill in the blanks

1. With σ = 10 and n = 25, the standard error of the mean is ____. **[2]**
2. $\bar{x}$ = 52, μ_0 = 50 and SE = 1 give Z = ____, so a two-tailed test at 5% ____ H_0. **[2; rejects]**
3. With P_0 = 0.5, n = 100 and p = 0.6, the Z statistic is ____. **[2]**
4. Making the sample four times larger makes the standard error ____. **[half as large]**
5. Z = 1.8 in a right-tailed test at 5% means H_0 is ____. **[rejected (1.8 > 1.645)]**
6. Samples of 100 with p_1 = 0.4 and p_2 = 0.6 give a pooled proportion $\hat{P}$ = ____. **[0.5]**